

EV Power - Lab 4 Project Report

Example Solution 1

Part 0: libraries

```
library(stringr)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggplot2)
library(tidyr)
library(maps)
library(sf)
```

Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE

```
library(readr)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v forcats 1.0.1      v purrr 1.1.0
v lubridate 1.9.4    v tibble 3.3.0

-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
x purrr::map()     masks maps::map()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

Part 1: Defining Research Question I want to find which state has the cheapest energy on average (i.e. how much energy per dollar per state from 2021 to 2023, energy is measured in BTU)

Research Question: ## **Part 2: Data Preparation and Cleaning**

```
price_data_21to23 <- read.csv("av-energy-price-2021-2023.csv")
total_energy_2021 <- read.csv("total-use-2021.csv")
total_energy_2022 <- read.csv("total-use-2022.csv")
total_energy_2023 <- read.csv("total-use-2023.csv")
price_vec <- unlist(price_data_21to23)

price_vec <- str_remove(price_vec, "Total\\.energy\\.average\\.price\\.\\.dollars\\.per\\.mi")

price_vec = str_split(price_vec, ",")

price_vec<- price_vec[-1]

price_tbl <- price_vec %>%
  map(~as_tibble_row(set_names(.x, c("State", "2021", "2022", "2023")))) %>%
  bind_rows()

price_tbl<- price_tbl|> slice(-1)

price_tbl_clean <- price_tbl |>
  mutate(
    `2021` = as.numeric(str_extract(`2021`, "\\d+\\.?\d*")),
    `2022` = as.numeric(str_extract(`2022`, "\\d+\\.?\d*")),
    `2023` = as.numeric(str_extract(`2023`, "\\d+\\.?\d*"))
```

```

)

price_2021 <- select(price_tbl_clean, State, `2021`)
price_2022 <- select(price_tbl_clean, State, `2022`)
price_2023 <- select(price_tbl_clean, State, `2023`)

energy_2023 = as_tibble(total_energy_2023)
energy_2022 = as_tibble(total_energy_2022)
energy_2021 = as_tibble(total_energy_2021)

```

Part 3: Joining / Pivoting Datasets for Analysis

```

energy_2021_long <- energy_2021 |> pivot_longer(
  cols = -Energy_Source,
  names_to = "State",
  values_to = "Energy_Use")

energy_2022_long <- energy_2022 |>
  pivot_longer(
    cols = -Energy_Source,
    names_to = "State",
    values_to = "Energy_Use")

energy_2023_long <- energy_2023 |>
  pivot_longer(
    cols = -Energy_Source,
    names_to = "State",
    values_to = "Energy_Use"
  )

avg_energy_2021 <- energy_2021_long |>
  group_by(State) |>
  summarize(avg_energy_use = mean(Energy_Use))

avg_energy_2022 <- energy_2022_long |>
  group_by(State) |>
  summarize(avg_energy_use = mean(Energy_Use))

avg_energy_2023 <- energy_2023_long |>

```

```

group_by(State) |>
  summarize(avg_energy_use = mean(Energy_Use))

use_v_price_2021 = left_join(avg_energy_2021, price_2021)

```

Joining with `by = join\_by(State)`

```
use_v_price_2022 = left_join(avg_energy_2022, price_2022)
```

Joining with `by = join\_by(State)`

```
use_v_price_2023 = left_join(avg_energy_2023, price_2023)
```

Joining with `by = join\_by(State)`

```
head(use_v_price_2021)
```

```

# A tibble: 6 x 3
  State avg_energy_use `2021`
  <chr>      <dbl>   <dbl>
1 AK          136995    20.0
2 AL          470531.   17.8
3 AR          227205    18.4
4 AZ          336251.   25.1
5 CA         1228450.   28.4
6 CO          272831    20.6

```

```
head(use_v_price_2022)
```

```

# A tibble: 6 x 3
  State avg_energy_use `2022`
  <chr>      <dbl>   <dbl>
1 AK          146055.   27.3
2 AL          467503.   23.4
3 AR          235623    23.8
4 AZ          330371.   31.7
5 CA         1248835.   37.4
6 CO          282295.   25.8

```

```
head(use_v_price_2023)
```

```
# A tibble: 6 x 3
  State avg_energy_use `2023`
  <chr>      <dbl>   <dbl>
1 AK          149396.    23.8
2 AL          453002.    21.1
3 AR          230212.    21.8
4 AZ          342533.    30.3
5 CA         1285964.    35.7
6 CO          271901.    23.8
```

Part 4: Mapping Visualization

```
states_map <- map_data("state")

use_v_price_2021 <- use_v_price_2021 |>
  mutate(State_full = tolower(state.name[match(State, state.abb)]))

use_v_price_2022 <- use_v_price_2022 |>
  mutate(State_full = tolower(state.name[match(State, state.abb)]))

use_v_price_2023 <- use_v_price_2023 |>
  mutate(State_full = tolower(state.name[match(State, state.abb)]))

# calculate the average consumption in thousand Btu, per dollar
btu_per_dollar_2021 <- use_v_price_2021 |>
  mutate(btu_per_dollar = avg_energy_use/`2021`)
btu_per_dollar_2022 <- use_v_price_2022 |>
  mutate(btu_per_dollar = avg_energy_use/`2022`)
btu_per_dollar_2023 <- use_v_price_2023 |>
  mutate(btu_per_dollar = avg_energy_use/`2023`)

map_2021 <- left_join(states_map, btu_per_dollar_2021, by = c("region" = "State_full"))
map_2022 <- left_join(states_map, btu_per_dollar_2022, by = c("region" = "State_full"))
map_2023 <- left_join(states_map, btu_per_dollar_2023, by = c("region" = "State_full"))
btuperdollar_map_2021 <- ggplot(map_2021, aes(x = long, y = lat, group = group, fill = btu_p
```

```

geom_polygon(color = "white", size = 0.2) +
coord_fixed(1.3) +
scale_fill_gradient(low = "lightblue", high = "darkred", na.value = "grey90") +
labs(
  title = "BTU per Dollar by State (2021)",
  fill = "BTU per $"
) + theme_void() + theme(
  legend.position = "bottom",
  legend.key.width = unit(2.5, "cm"),
  plot.title = element_text(hjust = 0.5, margin = margin(b = 15))
)

```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
 i Please use `linewidth` instead.

```

btuperdollar_map_2022 <- ggplot(map_2022, aes(x = long, y = lat, group = group, fill = btu_p
geom_polygon(color = "white", size = 0.2) +
coord_fixed(1.3) +
scale_fill_gradient(low = "lightblue", high = "darkred", na.value = "grey90") +
labs(
  title = "BTU per Dollar by State (2022)",
  fill = "BTU per $"
) + theme_void() + theme(
  legend.position = "bottom",
  legend.key.width = unit(2.5, "cm"),
  plot.title = element_text(hjust = 0.5, margin = margin(b = 15))
)

```

```

btuperdollar_map_2023 <- ggplot(map_2023, aes(x = long, y = lat, group = group, fill = btu_p
geom_polygon(color = "white", size = 0.2) +
coord_fixed(1.3) +
scale_fill_gradient(low = "lightblue", high = "darkred", na.value = "grey90") +
labs(
  title = "BTU per Dollar by State (2023)",
  fill = "BTU per $"
) + theme_void() + theme(
  legend.position = "bottom",
  legend.key.width = unit(2.5, "cm"),
  plot.title = element_text(hjust = 0.5, margin = margin(b = 15))
)

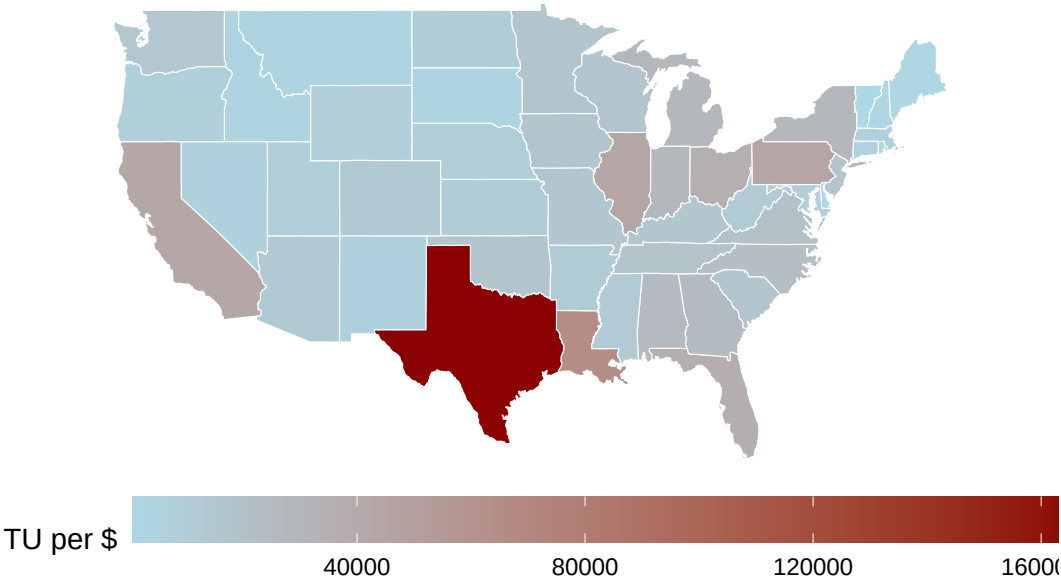
```

```

btuperdollar_map_2021

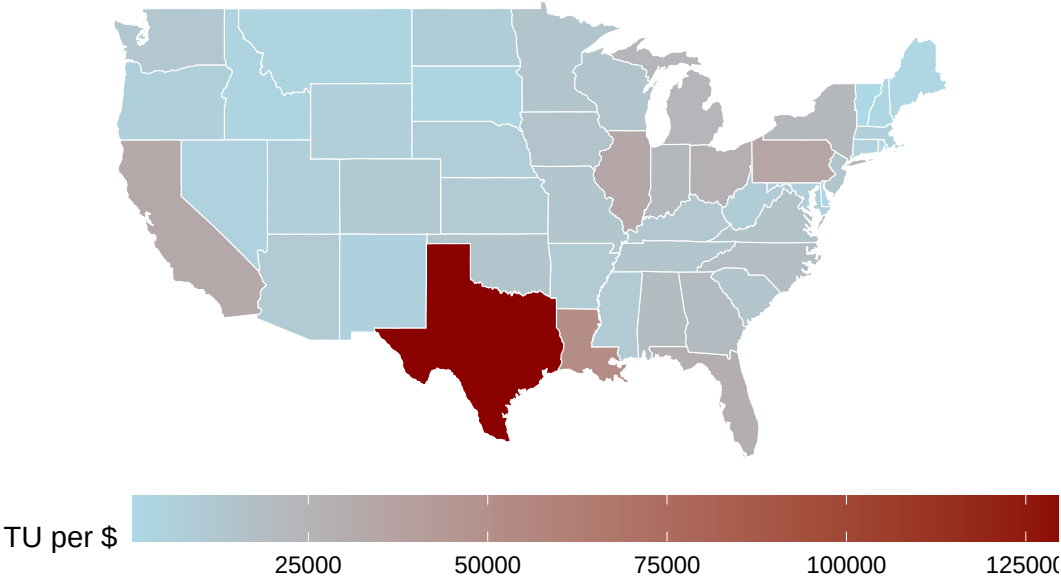
```

BTU per Dollar by State (2021)



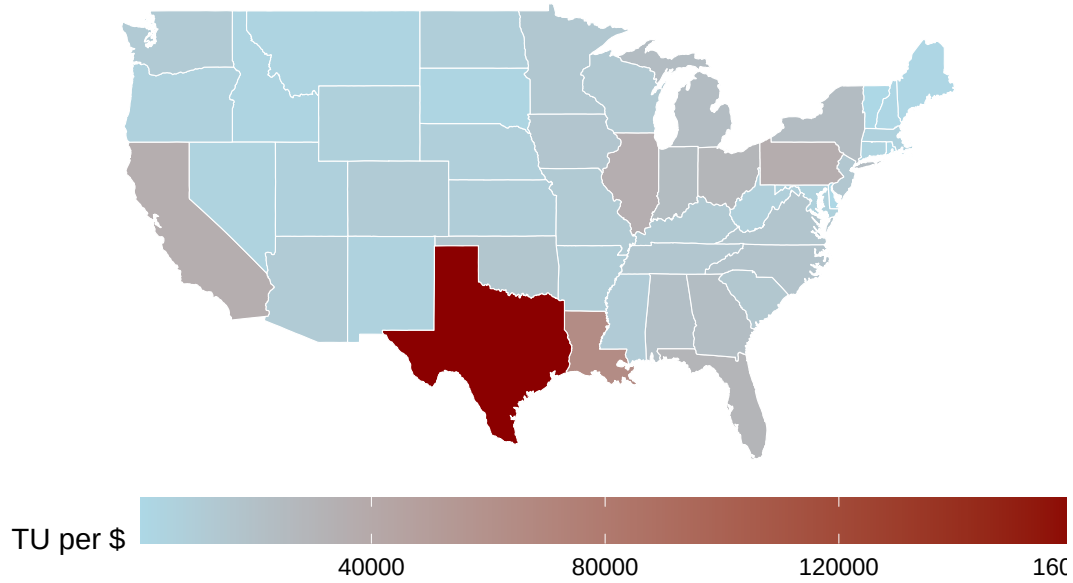
btuperdollar_map_2022

BTU per Dollar by State (2022)



```
btuperdollar_map_2023
```

BTU per Dollar by State (2023)



#Texas by far has the greatest amount of energy use per dollar, meaning that it is the cheapest